**GAME JAM PRODUCT PACK**

# LAST LIGHT

## *GAME DESIGN DOCUMENT*

Mobile, one-finger 2D pixel-art game design

***“How much of what you are trying to save can you sacrifice to survive?”***

| | |
|---|---|
| **Document version** | v1.0 - English Edition |
| **Date** | 17 July 2026 |
| **Product context** | Unity, solo developer, 48 hours, AI-assisted development |
| **Target platform** | Mobile-first, portrait 9:16; WebGL/Windows secondary |
| **Original working title** | Son Isik |

| STATUS | PURPOSE | PRIMARY AUDIENCE |
| --- | --- | --- |
| Planning input | Define gameplay, content, UX, technical boundaries, and production scope. | Product Manager, developer, and playtest participants |

## Contents

# 1. Product Summary and Design Principles

## 1.1 High concept

Last Light is a mobile avoidance and decision game built around 3-5 minute runs. The player carries a relic that must reach someone close to death. The traveller advances automatically while the player dodges obstacles with one-finger horizontal movement. At the end of each stage, the player either pays a permanent cost to buy safety or keeps everything and accepts a more dangerous next stage.

> **THEMATIC PROMISE**
> The theme will not be delivered only through text. Every advantage requires the player to give up part of the character, their senses, or the light they are trying to save.

## 1.2 Design goals

- One-finger mobile controls understood within the first 10 seconds.
- Every choice visibly and tangibly changes the next part of play.
- Short runs that support immediate retry and experimentation with different sacrifices.
- A small set of reusable systems and content suitable for a 48-hour solo build.
- Strong silhouettes, high contrast, and low asset cost through minimal 2D pixel art.

## 1.3 Experience pillars

| Pillar | Intended player feeling | Design response |
|---|---|---|
| Immediate readability | Understand control and danger without explanation | Portrait view, large silhouettes, horizontal movement only, shape and color coding |
| Meaningful loss | Feel that a choice removes something real | Reduced vision, lower speed, fading light, loss of HUD information |
| Voluntary risk | Treat refusal to pay as a skill claim | Pay the cost / accept the risk decision pair |
| Short regret loop | Want to retry with different choices after the ending | 3-5 minute run, fast restart, decision recap |

## 1.4 Target player and use context

- Players sampling multiple game jam entries in a short period.
- Mobile users playing one-handed, without headphones, or in a short session.
- Players interested in decisions, atmosphere, and replay rather than mechanical complexity.
- Players who expect to restart within five seconds after failure.

# 2. Player Experience and Core Loop

## 2.1 Player fantasy

The player is not a powerful hero. They are a small, fragile traveller carrying the last light through a dark passage. The run begins with the character feeling agile and whole. As it continues, the character accumulates the physical and sensory traces of the player’s choices. The ending evaluates not only whether the destination was reached, but what was consumed along the way.

## 2.2 Core loop

1. The traveller advances automatically.
2. The player drags left and right to avoid obstacles.
3. Two altars or gates appear at the end of a stage.
4. The player pays a cost to buy safety, or pays nothing and increases risk.
5. The permanent effect is applied to the character, HUD, or next stage.
6. After six stages, the remaining light resolves the ending.
7. A decision recap appears and the player can restart immediately.

> **RUN-TIME TARGET**
> 3-5 minutes including introduction and ending. Each play segment lasts 20-30 seconds; each decision should take 3-6 seconds.

### 2.3 Success and failure states

| State | Outcome | Player feedback |
|---|---|---|
| Health reaches 0 | Run ends | Break effect, stage reached, light remaining, Retry |
| Stage completed | Enter decision space | World slows; two offers become readable |
| Stage 6 completed | Ending evaluation | Ending resolved from light level and paid costs |
| New run | All temporary state resets | One-tap start; no long loading or menu |

## 3. Controls, Camera, and Game Systems

### 3.1 Control scheme

| Input | Behaviour | Design note |
|---|---|---|
| Horizontal drag | Character eases toward the target X position | Finger does not need to be on the character |
| Hold position | Character keeps its horizontal position | Automatic forward motion continues |
| Release finger | Character stays in place | Sliding and inertia are outside MVP |
| Drag into altar zone | Selects the offer | No confirmation button |

Mouse drag uses the same interface for Editor and WebGL testing. Virtual joysticks, attack buttons, and multi-touch are explicitly out of scope.

### 3.2 Camera and screen

- Portrait 9:16 with a 360 x 640 reference resolution.
- Pixel Perfect Camera; Point filtering and no sprite compression.
- The traveller stays near the lower centre while the world moves downward.
- The lower screen remains visually quiet for input; threats are introduced in the upper half.
- The vision cost narrows playable visibility through a vignette or mask rather than camera zoom.

### 3.3 Body system

| Parameter | Start | Change | Limit / rule |
|---|---|---|---|
| Health | 3 | Obstacle contact -1; some costs heal | Run ends at 0 |
| Maximum health | 3 | Blood cost -1 | Minimum 1 |
| Horizontal speed | 100% | Speed cost about -15% | Minimum 60% |
| Character size | 100% | Body cost +12-15% | Maximum 145% |
| Damage invulnerability | 0 sec | 0.8-1.0 sec after damage | Prevents repeated contact damage |

### 3.4 Relic light system

The light is both the entrusted relic and the quality of the ending. It begins at 100%. Light costs create immediate safety but worsen the ending. A run may continue at 0%; this creates the darkest result, in which the traveller survives but has lost the purpose of the journey.

| Light | Ending category | Narrative result |
|---|---|---|
| 70-100% | Good ending | The traveller bears the cost; the relic reaches the destination with enough power. |
| 30-69% | Bittersweet ending | The target is saved, but the light is incomplete or short-lived. |
| 1-29% | Empty victory | The traveller arrives, but the relic can no longer fulfil its purpose. |
| 0% | Dark ending | The traveller survives with no light left to deliver. |

### 3.5 Cost and risk choice

Each decision point should compare one “buy safety” offer with one “pay nothing and increase danger” offer. This avoids forcing a choice between two punishments and gives skilled players permission to gamble on their ability.

**FAIRNESS RULE**
The heavier the cost, the clearer the benefit must be. Effects are not hidden or deliberately vague. Exact ending thresholds may remain undisclosed.

### 3.6 Difficulty modifiers

| Modifier | Effect | Implementation boundary |
|---|---|---|
| Flow speed | World and obstacles scroll faster | Cumulative cap around +35% |
| Obstacle density | Enable an additional obstacle prefab | A valid passage must remain |
| Corridor width | Narrow the safe route | Must remain possible at maximum character size |
| Shadow pressure | Shorter warning or extra strike | Minimum readable warning: 0.6 sec |

## 4. Level and Content Design

### 4.1 Obstacle catalogue

| Obstacle | Behaviour | Readability | Variation |
|---|---|---|---|
| Spike wall | Static block with a clear corridor | Red, sharp silhouette | Gap position and width |
| Moving blade | Moves back and forth horizontally | Spin animation and motion trail | Speed, phase, paired blades |
| Hunting shadow | Locks the player position, warns, then strikes | Target line or glowing floor | Warning time and repeat count |

### 4.2 Locked six-stage flow

| Stage | Learning goal | Obstacles | Example decision |
|---|---|---|---|
| 1 - Tutorial | Horizontal movement and collision | Widely spaced spikes | Heal / make next stage faster |
| 2 - Timing | Read a moving threat | Spikes + one blade | Slow stage / add second blade |
| 3 - Pressure | Respond to a warning | Spikes + shadow | Remove shadow / shorten warning |
| 4 - Attrition | Feel earlier sacrifices | Mixed layout | Gain shield / narrow corridor |
| 5 - Crisis | Moral tension between health and light | Dense mixed layout | Full heal / accelerate final stage |
| 6 - Final passage | Combine learned skills | All three obstacles | No decision; enter ending |

### 4.3 Content production method

The MVP will not use full procedural generation. Each stage is assembled from short, hand-authored obstacle chunks in a fixed sequence. If time remains, a stage may choose between two chunks in the same difficulty band.

- Obstacles are reused as prefabs with Inspector parameters.
- Every chunk must be passable at allowed minimum speed and maximum character size.
- All new mechanics are introduced by Stage 3; the last two stages only combine known elements.
- Damage never arrives from off-screen or without a readable warning.

# 5. Cost and Risk Catalogue

| Code | Offer | Benefit | Permanent cost / risk | MVP |
|---|---|---|---|---|
| C01 | Blood Cost | Restore all health | Maximum health -1 | Yes |
| C02 | Light Cost | Remove moving hazards next stage | Light -15% | Yes |
| C03 | Sight Cost | Next stage scrolls 25% slower | Vision permanently narrows | Yes |
| C04 | Body Cost | Negate the next hit | Character size +15% | Yes |
| C05 | Speed Cost | Reveal the safe route in advance | Horizontal speed -15% | Yes |
| C06 | Memory Cost | Reveal all hazards earlier | Hide health HUD | If time |
| C07 | Mercy Cost | Skip one stage | Light -25% | If time |
| C08 | Silence Cost | Remove shadow attacks | Disable one music layer | If time |
| R01 | Pay Nothing | No resource loss | Next stage speed +20% | Yes |
| R02 | Pay Nothing | No resource loss | Extra blade or narrower corridor | Yes |
| R03 | Pay Nothing | No resource loss | Shorter shadow warning | Yes |

**COST PERCEPTIBILITY TEST**
If the player cannot notice a change without checking the HUD, the cost is not strong enough. Numerical values are tuned through playtest; effect categories are locked.

# 6. UX/UI Flow and Example Experience

## 6.1 Screen flow

1. Opening: logo, “Bring the light to them,” and Tap to Start.
2. Micro-tutorial: left-right drag animation; disappears after first movement.
3. Gameplay HUD: health, light percentage, and Stage 1/6.
4. Decision space: gameplay slows; left and right offers appear as large cards.
5. Cost application: icon moves into the traveller and the related visual layer changes.
6. Death or ending: outcome text, decision recap, and Retry.

## 6.2 HUD layout

| Area | Content | Rule |
|---|---|---|
| Top left | Health icons | Display only; no touch target required |
| Top centre | Stage 3/6 | Small but high contrast |
| Top right | Light % | Primary meta resource; gold color and light icon |
| Lower centre | Traveller | Positioned so the finger does not fully cover it |
| Bottom 25% | Control area | No fixed button; keep mostly clear |

## 6.3 Decision-card content standard

Cards must be short enough to read while the game is slowing down. Each card contains a large icon, one benefit line, and a “Cost:” or “Risk:” label. Total card copy should ideally stay within 8-12 words.

| Good example | Avoid |
|---|---|
| HAZARDS SLOW DOWN<br>Cost: Your movement speed | In the next level the movement speed of enemies will be reduced by a specific amount, but... |
| RESTORE HEALTH<br>Cost: 15% of the light | The mysterious altar may choose to help you |
| PAY NOTHING<br>Risk: A second blade | Free option |

### 6.4 Example user experience

| Time | What the player sees | What the player thinks / does |
|---|---|---|
| 0:00-0:10 | “Bring the light to them”; drag gesture | Tests the control without reading a long tutorial. |
| 0:10-0:35 | Widely spaced spikes | Dodges left and right; one collision teaches feedback. |
| 0:35-0:45 | Heal by spending light, or pay nothing and speed up | Makes the first decision between personal safety and the relic. |
| 0:45-1:40 | Blade and shadow are introduced | Notices the consequences of earlier costs. |
| 1:40-2:50 | Vision, speed, or body has degraded | Adapts to a changed control condition. |
| 2:50-3:30 | Large light cost offered for a full heal | Hesitates between reaching the end and earning a better ending. |
| 3:30-4:00 | Ending and decision recap | Recognises a better result may be possible and restarts. |

### 6.5 Accessibility and mobile usability

- Color is never the only information channel; hazards also use shape and animation.
- Text uses a minimum readable size and high contrast.
- Haptics should be optional. Without a settings screen, use only brief impact vibrations.
- Shadow attacks remain fully understandable through visual warnings with audio muted.
- Notches and protected areas are handled through Canvas Safe Area support.

## 7. Visuals, Audio, and Feedback

### 7.1 Pixel-art direction

Use minimal dark pixel art. The world is built from deep indigo and navy, hazards are red, the traveller is pale gray or white, and the relic light is warm gold. The light remains the brightest element in the scene.

| Asset | Suggested scale | Animation scope |
|---|---|---|
| Traveller | 24x24 or 32x32 | 4 movement frames, 1 hit, 1 death |
| Light | 8x8-16x16 + glow | 2-4 flicker frames |
| Spikes | 16x16 tile | Static |
| Blade | 24x24 | 2-4 spin frames |
| Shadow | 24x24-32x32 | 2 warning frames + strike |
| Altar | 32x32 | 2-3 flame frames |
| Ending image | Simple silhouette | Static or 2 frames |

### 7.2 Visual traces of sacrifice

| Cost | Visual result | Production method |
|---|---|---|
| Maximum health | Cracks appear on the traveller | Toggleable overlay sprite |
| Speed | Chain attached to the foot | Child SpriteRenderer layer |
| Vision | Screen edges darken | UI mask / vignette |
| Body | Traveller and hitbox grow | Transform scale + controlled collider |
| Light | Carried light shrinks and fades | Scale + glow intensity |
| Memory | Health icons crack and disappear | HUD animation |

### 7.3 Audio and haptics

- Damage: short, weighty sound and light vibration.
- Cost selection: descending tone that emphasises loss.
- Spending light: bright sound that distorts or thins out.
- Stage transition: brief breath or closing gate.
- Music: one loop is sufficient; if time remains, remove layers as costs accumulate.
- Shadow: attack is readable visually even without sound.

# 8. Technical Design Boundaries

## 8.1 Recommended system separation

| System | Responsibility | Boundary |
|---|---|---|
| PlayerController | Touch/mouse horizontal movement | Contains no health or cost logic |
| PlayerHealth | Health, damage, invulnerability, death event | Updates UI through events |
| RelicLight | Light value and change events | Read by ending resolution |
| LevelFlow | Stage order, start/end, global speed | Does not own obstacle behaviour |
| CostData | Offer copy, icon, value, type | ScriptableObject |
| CostApplier | Apply cost or risk modifier | Keep switch/strategy simple |
| ChoiceGate | Present two offers and receive physical selection | Triggers only once |
| EndingResolver | Choose ending from light thresholds | Rule based |
| HUDController | Health, light, stage, and decision cards | Contains no game logic |

## 8.2 AI-assisted development rules

- Request small, single-responsibility scripts instead of generating the entire project.
- Every request includes setup steps, Inspector fields, and a test scenario.
- Do not rewrite existing class names or public APIs without explicit justification.
- Every system must be testable with the mouse in the Editor.
- AI output is not accepted until it compiles; null-reference and mobile-input tests are mandatory.
- No runtime generative AI, network dependency, or external API.

## 8.3 Performance targets

| Target | Criterion |
|---|---|
| Frame rate | Target 60 FPS on a mid-range Android device; 30 FPS hard floor |
| Loading | Less than 3 seconds from launch to play |
| Memory | No large texture atlases or video content |
| Resolution | Canvas Scaler + Safe Area for common 9:16 and 20:9 devices |
| Build | Android or WebGL primary submission; other platforms are bonus |

# 9. Scope, Acceptance Criteria, and Risks

## 9.1 MVP scope

| Must Have | Should Have | Could Have |
|---|---|---|
| One-finger movement<br>Automatic progression<br>2 obstacles<br>Health + light<br>Cost/risk choice<br>6 stages<br>2+ endings<br>Restart<br>Mobile build | Third obstacle<br>4-5 different costs<br>3-4 endings<br>Vision effect<br>Haptics<br>Decision recap<br>Basic music and SFX | Limited stage variation<br>Music layers<br>Memory/silence costs<br>Extra ending illustration<br>Achievement or score |

## 9.2 Explicitly out of scope

- Twin-stick combat, attack system, or boss.
- Inventory, skill tree, or meta progression.
- Full procedural level generation.
- Long dialogue, branching story system, or voice acting.
- Online services, leaderboards, or runtime AI.
- Multiple playable characters or environment biomes.

## 9.3 Product acceptance criteria

| Area | Acceptance criterion |
|---|---|
| Control | A first-time player can move left and right within 10 seconds without reading instructions. |
| Theme | At least three costs permanently and visibly change play. |
| Flow | The game can be completed from start to ending in under 5 minutes; restart takes under 5 seconds. |
| Fairness | Every stage is passable at allowed minimum speed and maximum hitbox. |
| Mobile | Input, safe area, and performance are verified on at least one real Android device. |
| Ending | At least two outcomes trigger correctly from remaining light. |
| Stability | No blocker bug, soft lock, or altar that can be selected twice in the main flow. |

### 9.4 Major risks and mitigation

| Risk | Impact | Mitigation |
|---|---|---|
| Cost combinations make the game impossible | Unfairness and abandonment | Test min/max combinations; cap modifiers |
| Pixel-art production expands | Core loop is delayed | Use placeholders for first 16 hours; layered sprites; limited asset list |
| Mobile input feels delayed or imprecise | Game becomes unplayable | Early real-device test; tune smoothing; reserve lower control area |
| Decision copy is not readable | Theme is missed | 8-12 word card standard; slow the game during decisions |
| AI-generated code damages architecture | Bugs and integration debt | Small tasks, stable public API, compile and smoke test every step |
| Feature growth | Build misses deadline | Cut list and feature freeze; protect Must Have scope |

## 10. Playtest Plan and Open Decisions

### 10.1 First playtest questions

- How many seconds did it take to understand the control?
- Could the player correctly describe what was gained and lost in the first decision?
- Did “pay nothing” feel like a real option?
- Which cost was most noticeable? Which one went unnoticed?
- Did death feel fair or caused by the player’s own mistake?
- After the ending, did the player want to retry? What would they change?
- Did the run feel too short, too long, or appropriate?

### 10.2 Playtest measures

| Measure | Target signal |
|---|---|
| Stage 1 completion rate | 85%+ |
| See an ending on first run | 40-70%; neither trivial nor excessively punishing |
| Average run time | 3-5 minutes |
| Correctly explain a cost | Most testers can state both the benefit and the loss |
| Intent to retry | At least half want to try different choices |
| Blocker / soft lock | 0 |

### 10.3 Decisions to confirm with the PM

| Topic | Default recommendation | Decision signal |
|---|---|---|
| Primary build | Android | Jam submission rules may shift priority to WebGL |
| Game title | Last Light (original: Son Isik) | Brand fit and itch.io discoverability |
| Number of endings | 3 main + 1 dark | Art capacity may reduce to 2-4 |
| Randomisation | Fixed stage sequence in MVP | Revisit after replay-intent playtest |
| Language | Short English copy and language-independent icons | Target jam audience |
| Telemetry | Manual playtest notes | Add simple event log only if platform allows |

**HANDOFF OUTCOME**
This document defines how the game behaves. Work sequence, hour estimates, ownership, and daily tracking should be created separately in schedule and task-management planning.